

5 . EXPLAINABILITY

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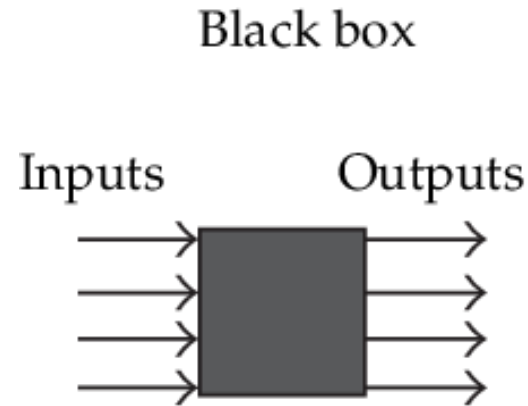


1 . I N T R O D U C T I O N

WHY EXPLAINABILITY?

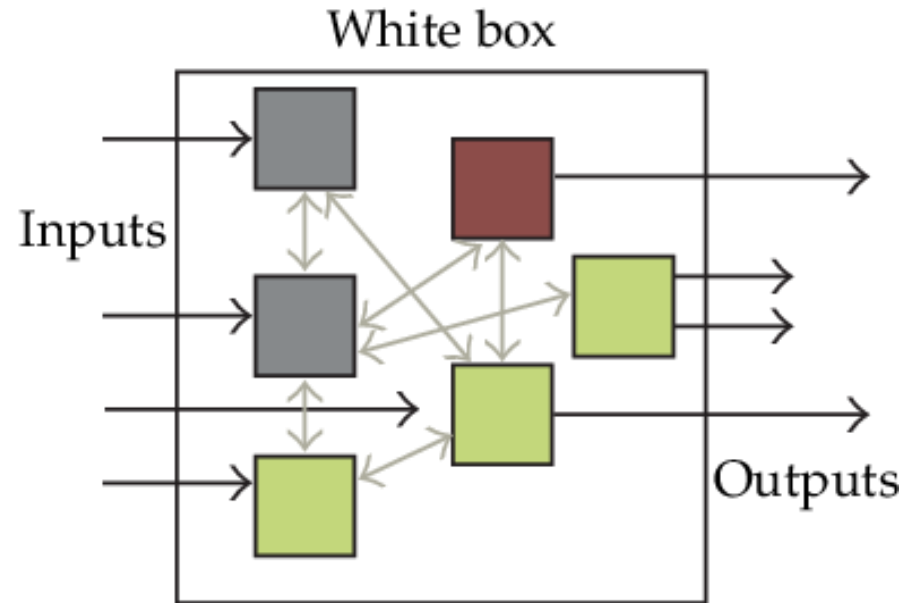


BLACK BOX VS WHITE BOX MODELS



- Neural Networks
- Ensembles
- ...

(Explainability)



- Linear Regression
- Decision Trees
- ...

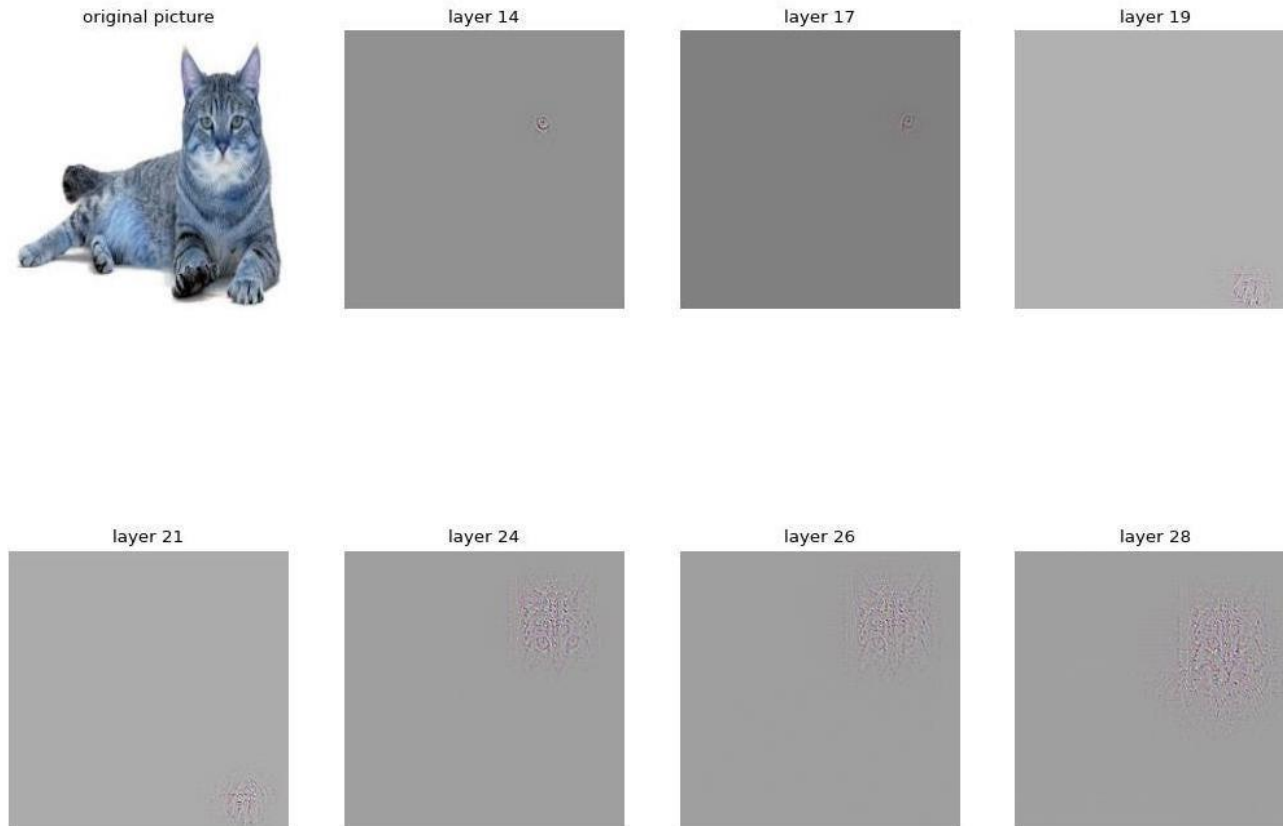
(Interpretability)

In this chapter, we will focus on generating explanations for black-box models (although the methods also work for white-box models).



2. VISUALIZATION OF INTERNAL FEATURES

MAXIMALLY ACTIVATING PATCHES

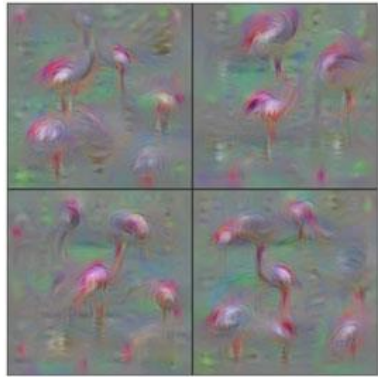


1. Run an image through the network and record the activations of chosen channel/layer.
2. Reshape the activations to the shape of the original image.
3. Visualize image patches that correspond to maximal activation.

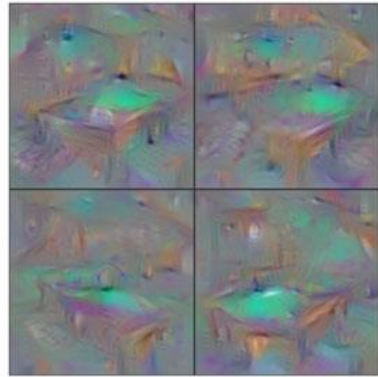
What types of (real) patterns cause certain layers or filters to activate?



GRADIENT ASCENT



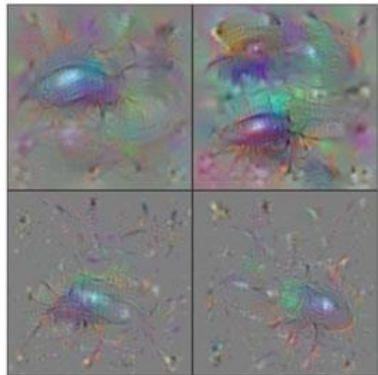
Flamingo



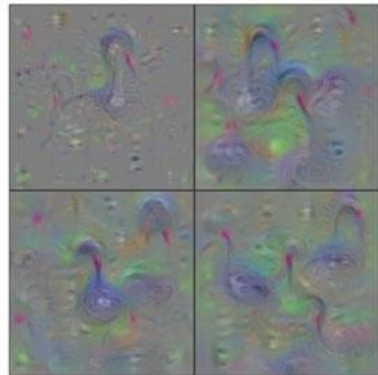
Billiard Table



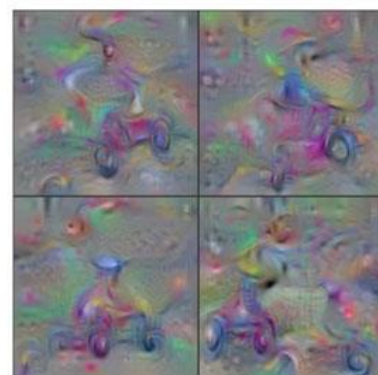
School Bus



Ground Beetle



Black Swan



Tricycle

$$\operatorname{argmax}_x \underbrace{\{\operatorname{agg}(f(x)) - \lambda ||x||_2\}}_{\mathcal{F}(x)}$$

$$x_{t+1} = x_t + \alpha \Delta \mathcal{F}(x_t)$$

What (synthetic) patterns activate a specific layer or filter?



FEATURE INVERSION



x

$$\operatorname{argmin}_x \{ ||f(x) - f(x_0)||_2^2 + \lambda ||x||_2 \}$$

We also train with gradient descent.



x

"What information is the network actually storing in its internal representations?"



x

- Shallower layers: similar reconstructions
 - Deeper layers: more fuzzy reconstructed images
- This shows what the network think it's relevant to keep.



DEEPDREAM



First layers



Last layers

Like gradient ascent, but starting with a real image.

What a neural network "sees" when asked to boost what it already recognizes?



3 . P E R T U R B A T I O N - B A S E D M E T H O D S

I D E A

- Hide parts of the input and observe how the output of the model changes.
- Model agnostic: we can apply this technique to any model.
- Intuitive and easy, but computationally expensive.



SHAP



1. Divide the image into regions.
2. All possible versions of the image are created where some regions are present and others are hidden.
3. Calculation of SHAP values.



SHAPLEY VALUES

$$N = \{1, \dots, M\} \text{ regions}$$

$$\underbrace{\phi_i}_{\text{SHAP value for region } i} = \sum_{\underbrace{S \subseteq N - \{i\}}_{\text{for every subset without } i}} \underbrace{\frac{|S|!(M - |S| - 1)!}{M!}}_{\text{fairness in the assignment}} \underbrace{|f(S \cup \{i\}) - f(S)|}_{\text{comparing when } i \text{ is also visible}}$$



EXAMPLE (SHAPLEY VALUES)

$$N = \{A, B, C\}$$

$$S \subseteq N - \{A\}: S_0 = \emptyset; S_1 = \{B\}; S_2 = \{C\}; S_3 = \{B, C\}$$

$$W(S_0) = 0!(3-0-1)! / 3! = 1/3 \quad (\text{A first})$$

$$W(S_1) = \dots = 1/6 \quad (\text{A second and B first})$$

$$W(S_2) = \dots = 1/6 \quad (\text{A second and C first})$$

$$W(S_3) = \dots = 1/3 \quad (\text{A third, first and second B-C or C-B})$$

A B C
A C B
B A C
B C A
C A B
C B A

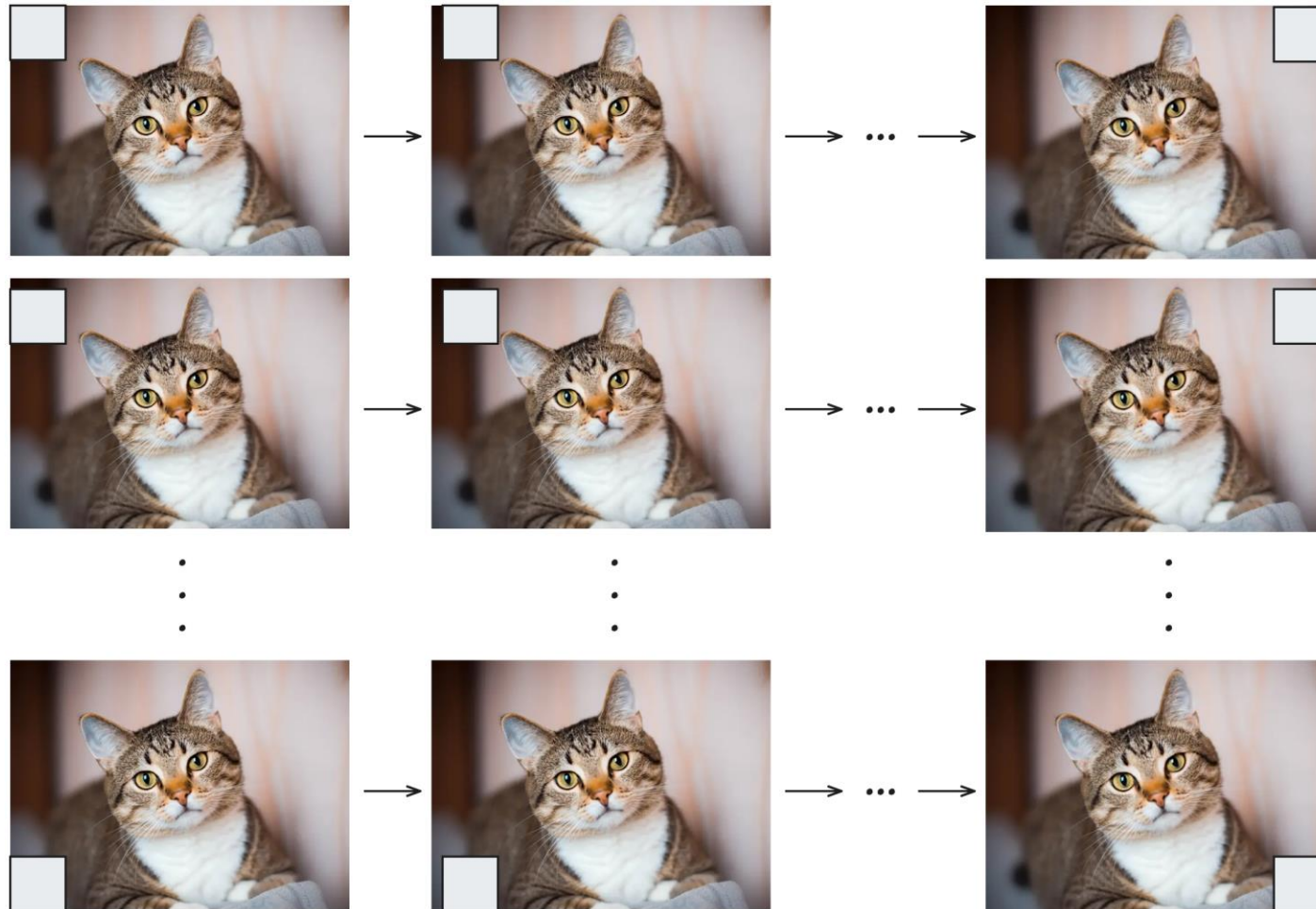


SHAP IN PRACTICE

1. Generate dataset with a sample of all possible subsets S .
2. Weight each subset with Shapley kernel (optional).
3. Train a linear regression model:
 - Inputs: One-hot vector indicating if region $i \in \{1, \dots, M\}$ is present.
 - Labels: Prediction of the original model.
4. Parameters of the linear regression are the Shapley values.



OCCCLUSION

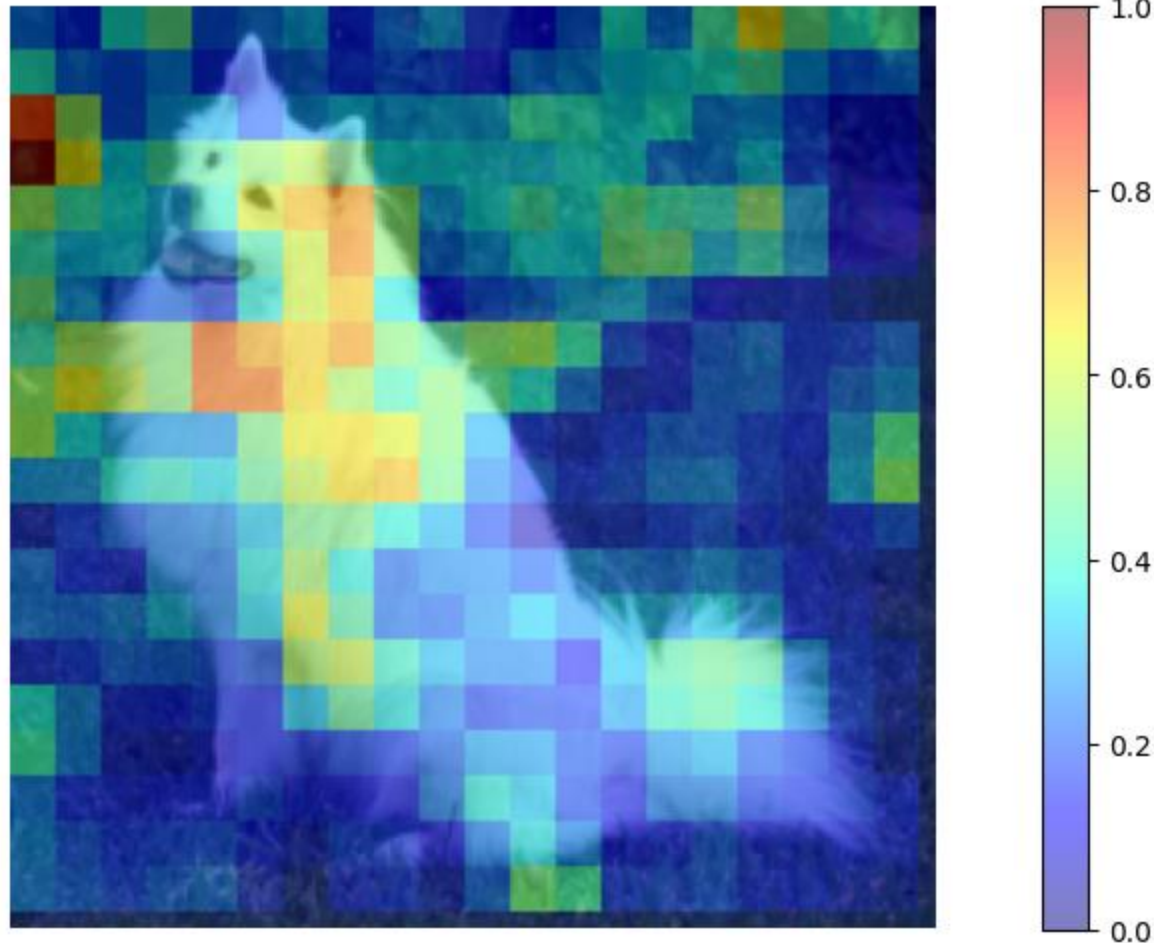


- We use one matrix to count where we are masking.
- We use another matrix to count the difference in the predictions.



OCCCLUSION

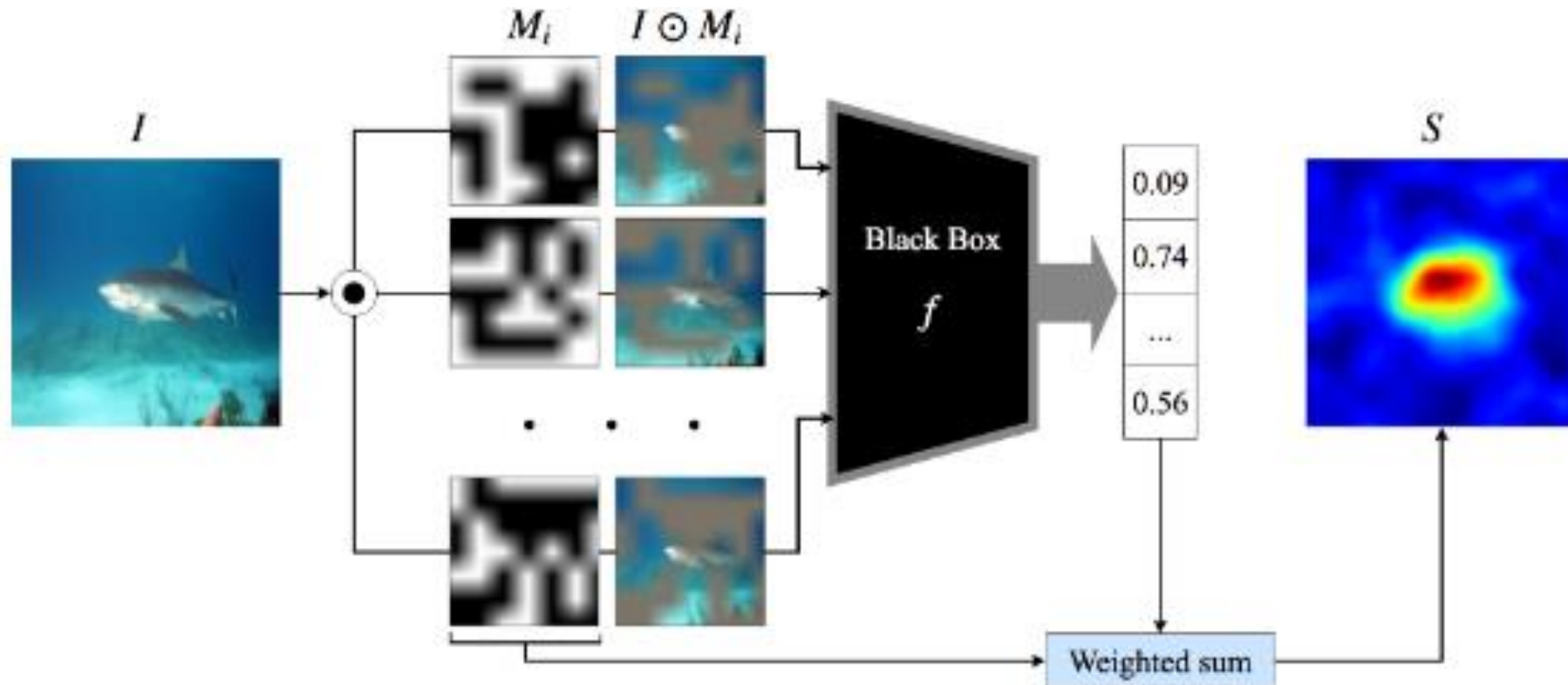
Occlusion



The more accurate the explanation (smaller patches), the more forward passes we need to do, so more computationally expensive.



RISE



- Like occlusion, but values are interpreted as contributions of the pixels rather than changes caused by masked pixels.
- We can generate masks in different ways, but a simple one is just using a Bernoulli distribution.

4 . GRADIENT-BASED METHODS

IDEA

- Measure the sensitivity of the output to each pixel in the image.
- Model specific: we can not apply this technique to any model (but may be to a lot).
- More complex, but less computationally expensive.



SALIENCY MAP

Idea: Calculate how sensitive the model's output is to each input pixel.

$$S_{ij}(x) = \max_{k \in \{1, 2, 3\}} \left| \frac{\partial f_c(x)}{\partial x_{kij}} \right|, \quad S \in \mathbb{R}^{H \times W}$$

- It can sometimes be noisy.
- No sense of direction.



INPUT X GRADIENT

Idea: If a pixel has a high value and the gradient is also high, then that pixel has a strong influence on the prediction.

$$S_{ij}(x) = \max_{k \in \{1, 2, 3\}} \left| x_{kij} \frac{\partial f_c(x)}{\partial x_{kij}} \right|, \quad S \in \mathbb{R}^{H \times W}$$

- Same problems as the saliency map.
- Sometimes it can be more informative.
- Other times it can penalize small input values.



GUIDED BACKPROPAGATION

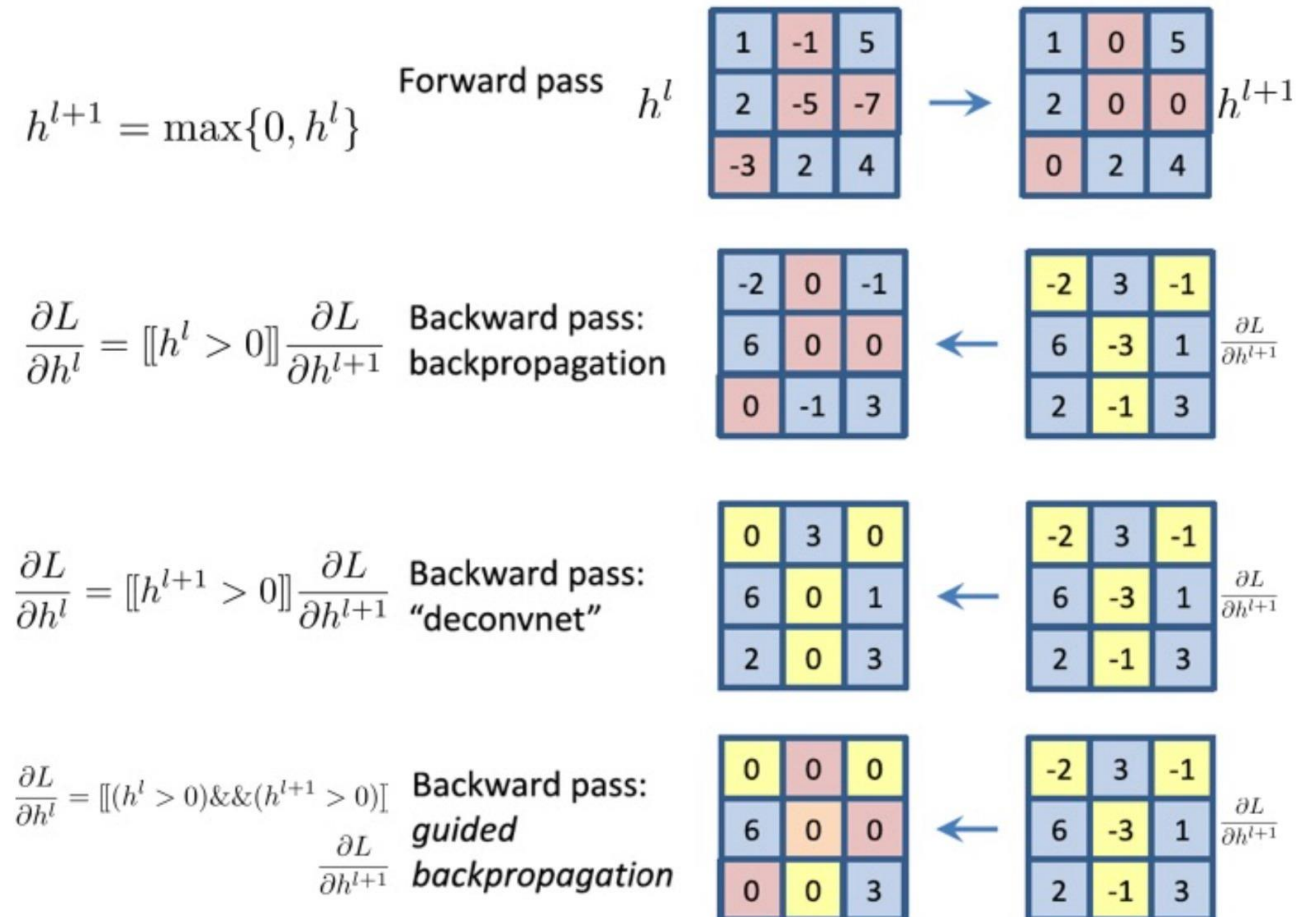
Idea: Modify the backpropagation process to block negative gradients in the ReLU layers during activation and backpropagation.

- Cleaner, more focused saliency maps.
- It does not respect the causal structure of the model.



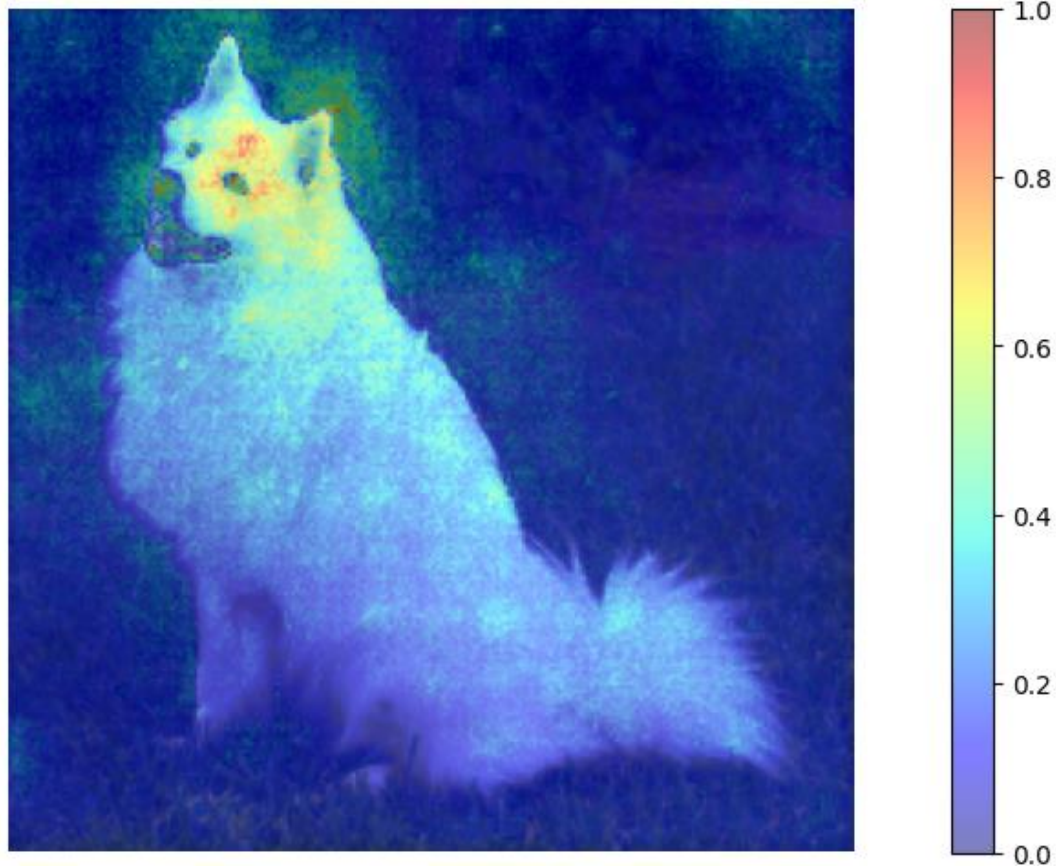
DECONVNET

Idea: Filter only the incoming gradient to visualize which input patterns excite an internal neuron.



SMOOTHGRAD

Idea: Add noise to remove noise!



$$x_i = x + \mathcal{N}(0, \sigma^2)$$

- Reduces noise averaging predictions.
- It can be applied to any other explanation method.
- More computationally expensive.



INTEGRATED GRADIENTS

Idea: Integrate gradients along a path from a base reference (e.g., a black image) to the original image.

- Reduces noise.
- It can be applied to any other explanation method.
- More computationally expensive.



INTEGRATED GRADIENTS

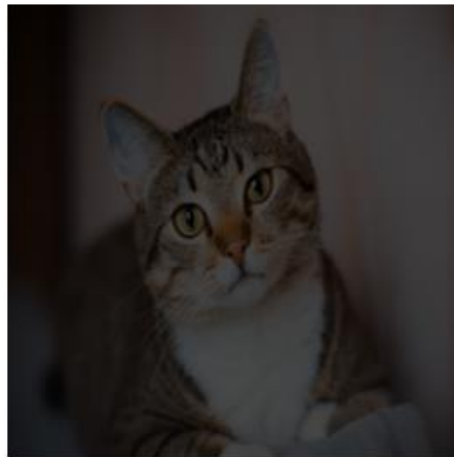
$$\text{IG}(x) = (x-x') \int_{\alpha=0}^1 \frac{\partial f(x' + \alpha(x-x'))}{\partial x} d\alpha \quad \underbrace{=} \quad (x-x') \lim_{M \rightarrow \infty} \frac{1}{M} \sum_{k=1}^M \frac{\partial f\left(x' + \frac{k}{M}(x-x')\right)}{\partial x}$$

integral is just an infinite sum!

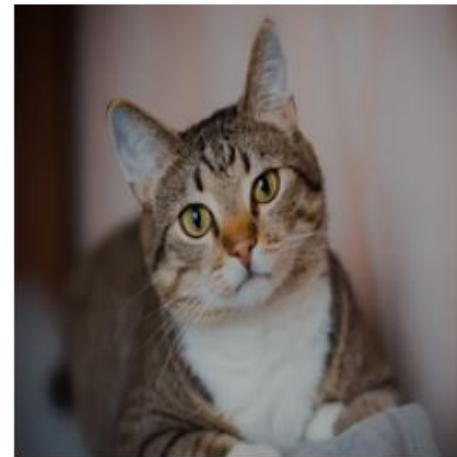
$\alpha = 0.00$



$\alpha = 0.33$



$\alpha = 0.66$



$\alpha = 1.00$



5 . A D - H O C M E T H O D S

IDEA

- Exploit the internal structure of the model.
- More model specific even than gradient-based methods.
- More complex, but less computationally expensive.



CLASS ACTIVATION MAPPING (CAM)

Idea: Where does the network look to classify an image as class c ?

We need:

- Global Average Pooling (GAP)

$$F_k(x) = \frac{1}{hw} \sum_{i=1}^h \sum_{j=1}^w A_{kij}(x) \quad \forall k = 1, \dots, n$$

number of channels of the layer

- And then a linear classification layer

$$f_c(x) = \sum_{k=1}^n w_k^c F_k(x)$$

Doing some math, we obtain

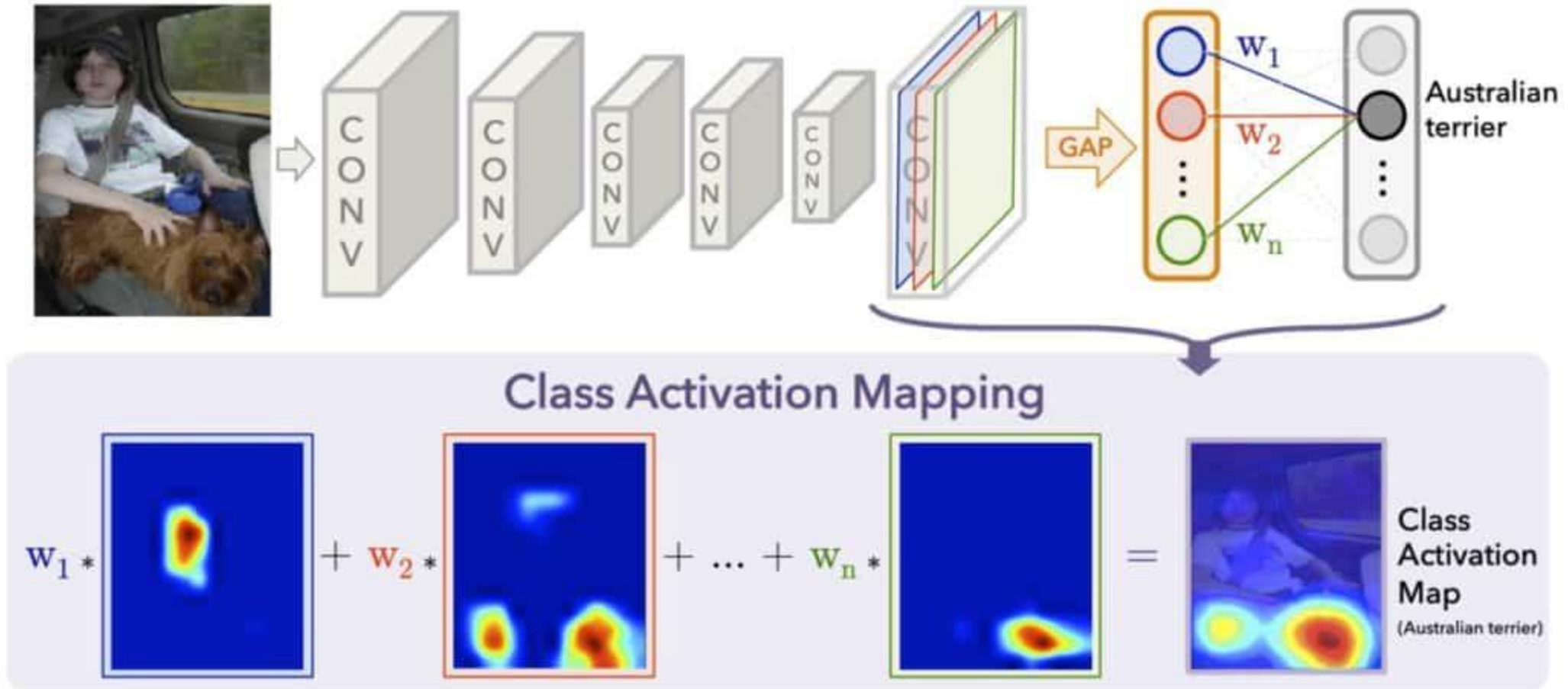
$$f_c(x) = \frac{1}{hw} \sum_{i=1}^h \sum_{j=1}^w \underbrace{\sum_{k=1}^n w_k^c A_{kij}(x)}$$

Finally, resize CAM to the shape of the original image.

Class Activation Map $\in \mathbb{R}^{h \times w}$



CLASS ACTIVATION MAPPING (CAM)



GRAD-CAM

Generalization of CAM (it can be applied to any convolutional network). number of channels of the layer

1. Choose a convolutional layer and calculate:

$$\frac{\partial f_c(x)}{\partial A_k(x)} \quad \forall k = 1, \dots, n$$

↑

2. Average over the spatial dimensions to obtain one alpha per channel:

$$\alpha_k^c = \frac{1}{hw} \sum_{i=1}^h \sum_{j=1}^w \frac{\partial f_c(x)}{\partial A_{kij}(x)}$$

3. Combine maps using these weights (alphas).
ReLU ensures that only regions positively associated with the class c are displayed.

$$M_c(i, j) = \text{ReLU} \left(\sum_{k=1}^n \alpha_k^c A_{kij} \right)$$

Finally, resize to the shape of the original image.



GUIDED GRAD-CAM

Idea: Combine Guided Backpropagation and Grad-CAM with element-wise multiplication.

- Guided Backpropagation tells you what pixels are important.
- Grad-CAM tells you which regions are important.
- Therefore, Guided Grad-CAM tells you which are the important pixels of the relevant regions.



ATTENTION AS EXPLAINABILITY

"Saying that attention is explainability is like saying that correlation is causation"



6 . EVALUATION

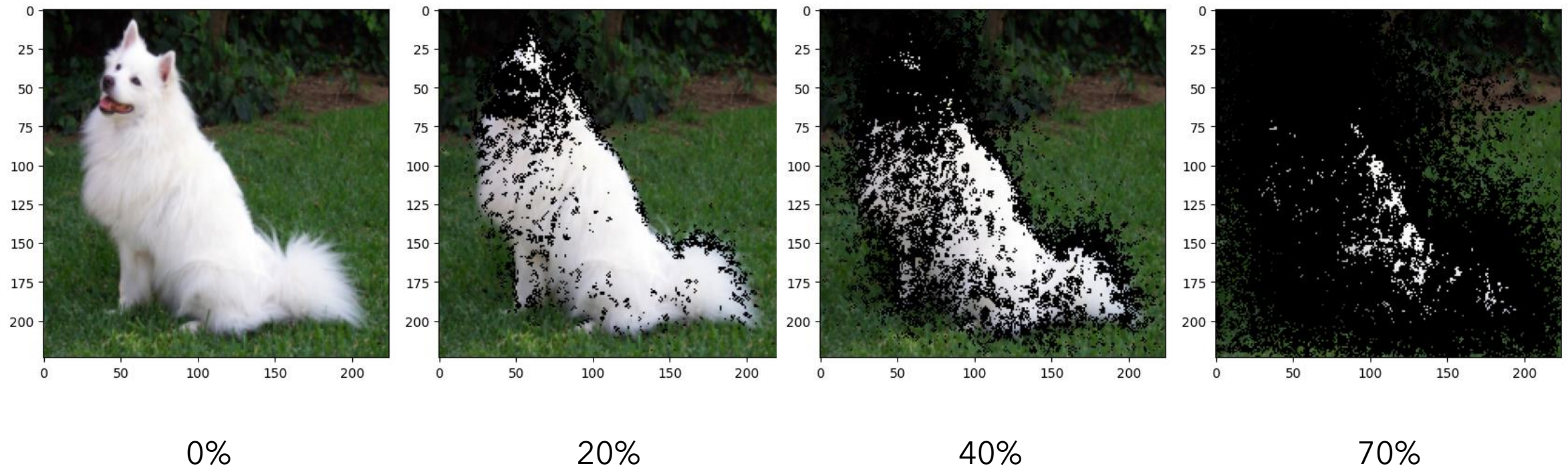
COMMON PROBLEMS

- Unstable gradients (can be solved with SmoothGrad).
- Rapid sign swings (can be solved taking absolute value).
- Noisy gradients (Guided Backpropagation can clean up gradients and Grad-CAM focus on the target class).
- Vanishing gradients (can be solved with Grad-CAM, as it can focus only on deeper layers).
- Lack of context/local sensitivity (can be solved with SmoothGrad, Integrated Gradients and Grad-CAM).



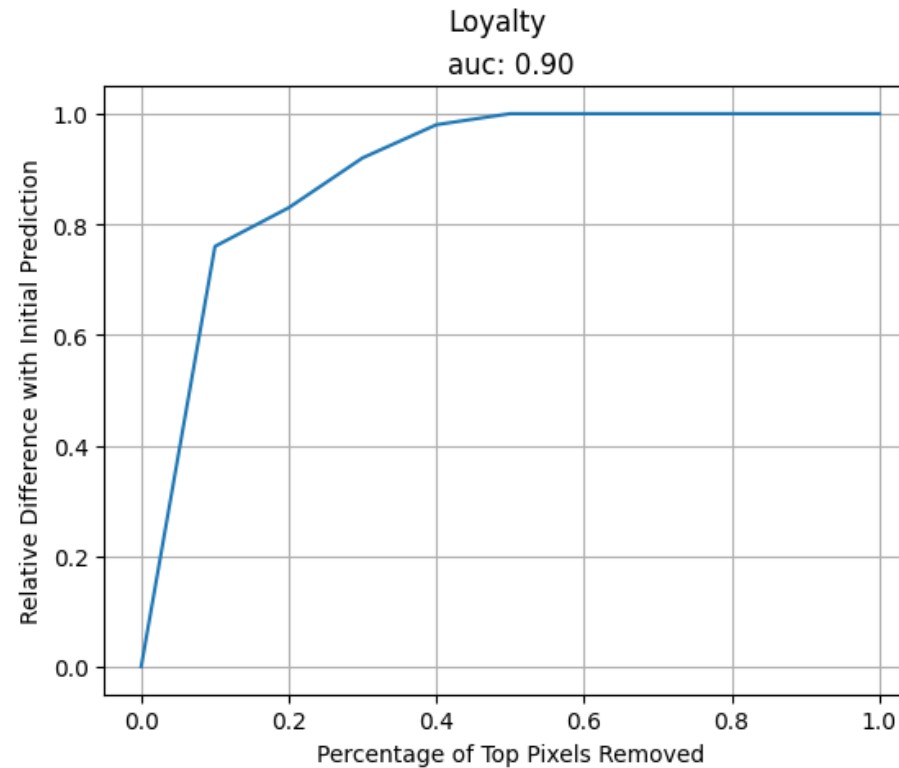
LOYALTY

How well the regions identified as important match what actually influences the model's prediction?



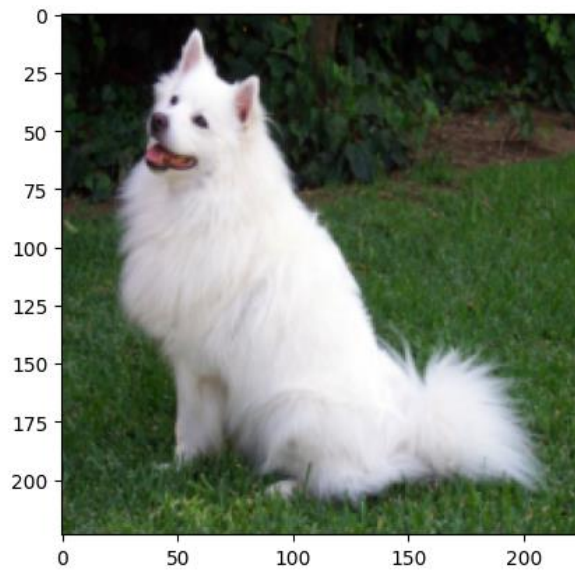
LOYALTY

- At first, it should increase heavily, because most important pixels are being removed.
- The closer to 1, the better the explanation.

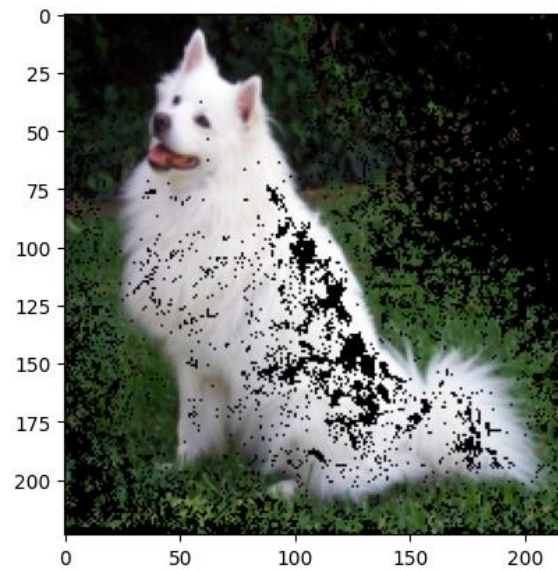


INVERSE LOYALTY

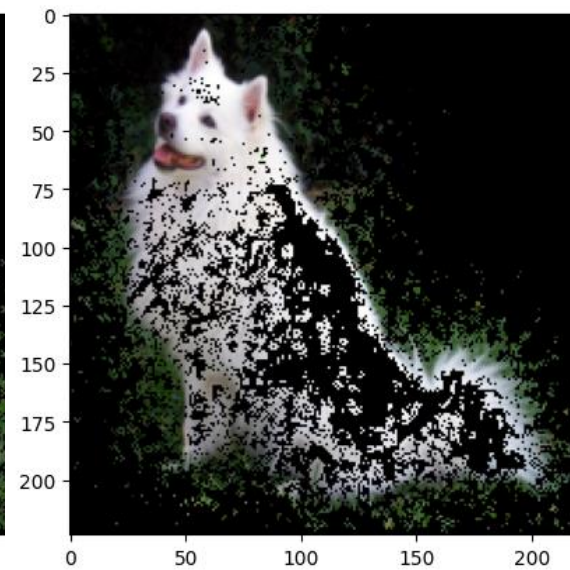
How well the regions identified as unimportant match what actually influences the model's prediction?



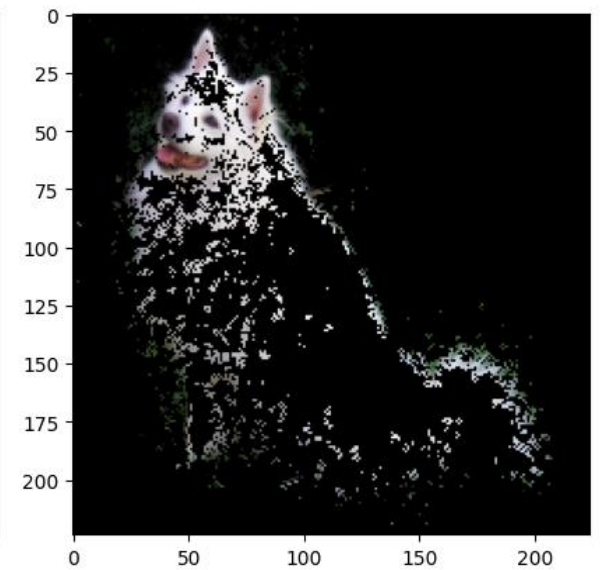
0%



20%



50%

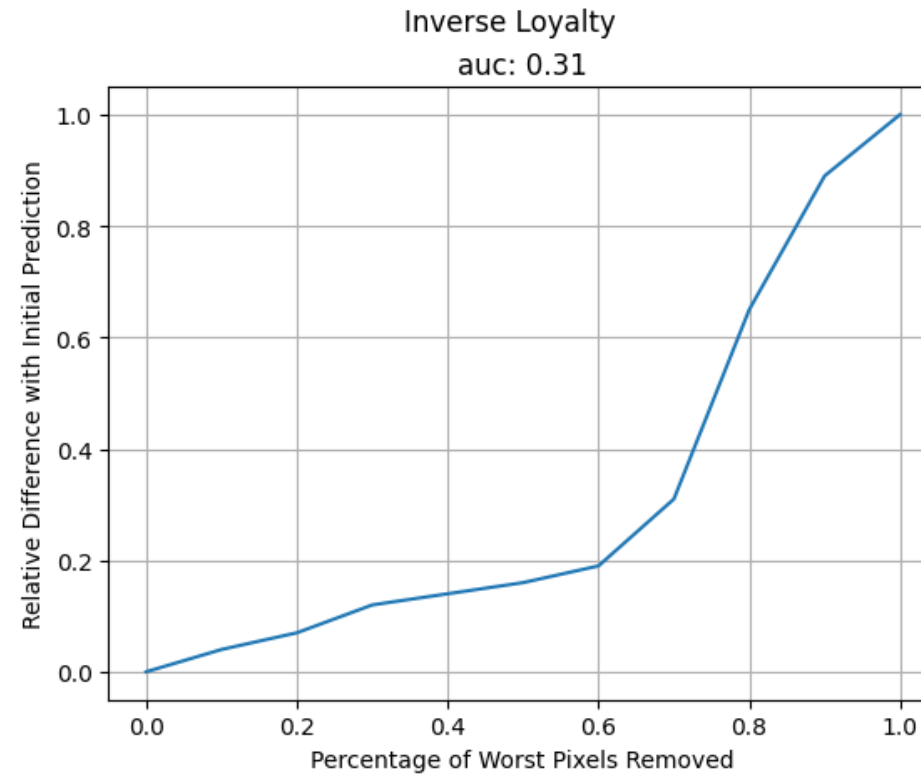


70%



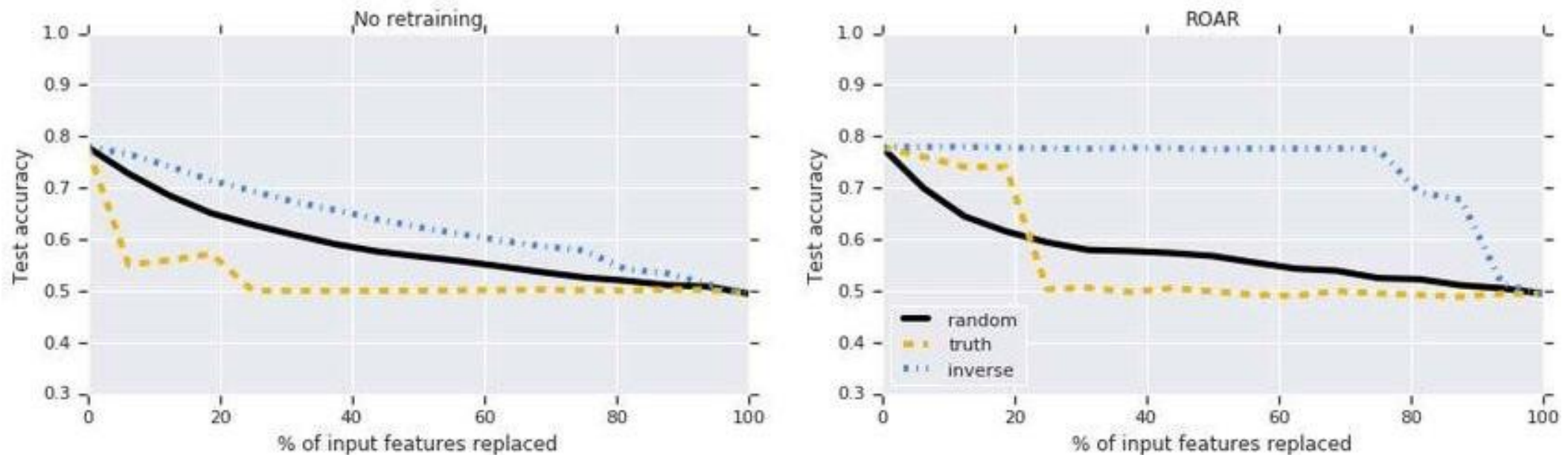
INVERSE LOYALTY

- At first, it should increase slowly , because less important pixels are being removed.
- The closer to 0, the better the explanation.



REMOVE AND RETRAIN (ROAR)

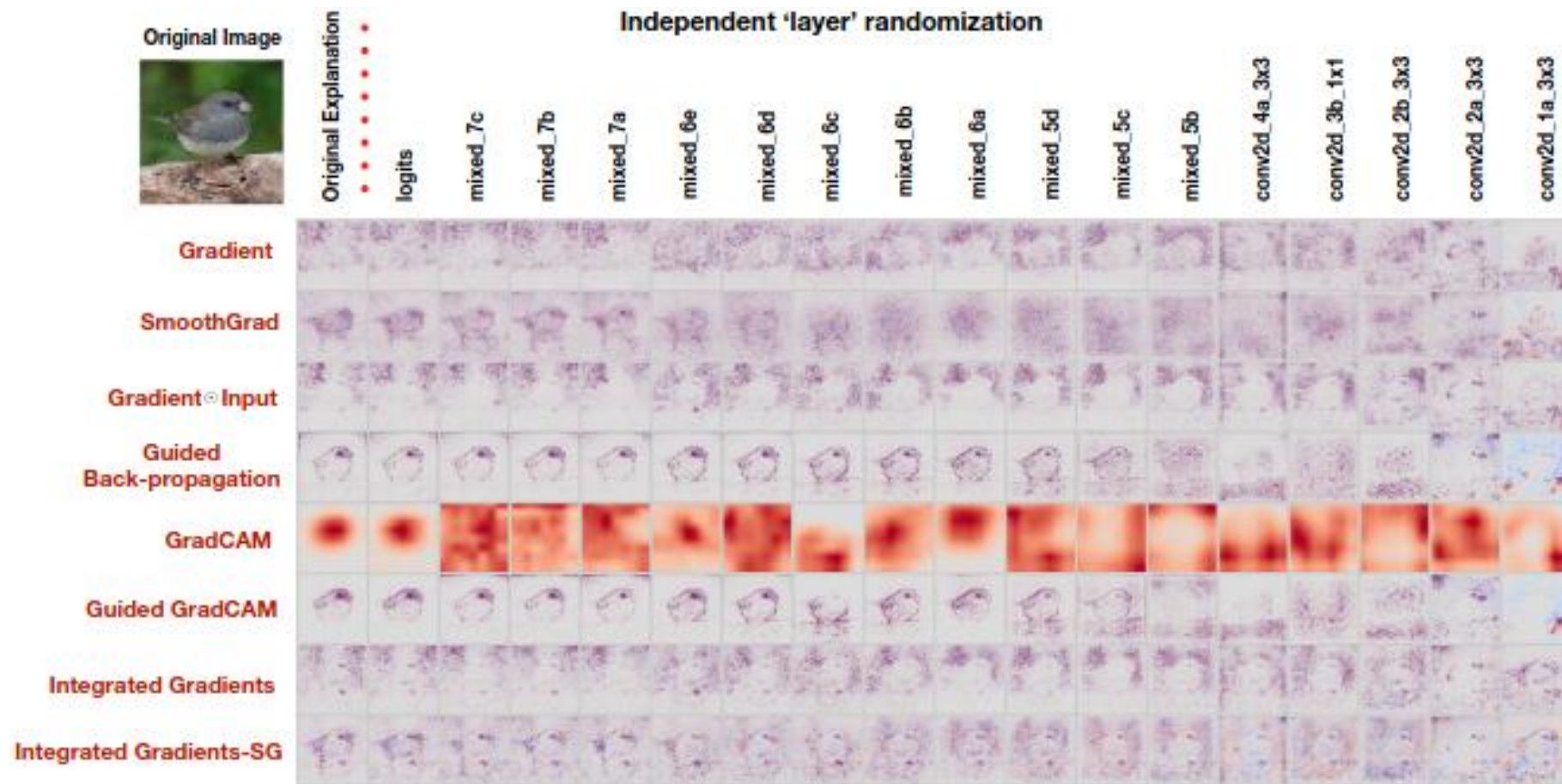
Idea: If removing what is "important" according to the explanation causes the model to suffer, then the explanation was useful in detecting what was relevant.



"La 'NOTTE' iba por 'QUI'... Y DE REPENTE... ¡¡POR 'QUA'!!"

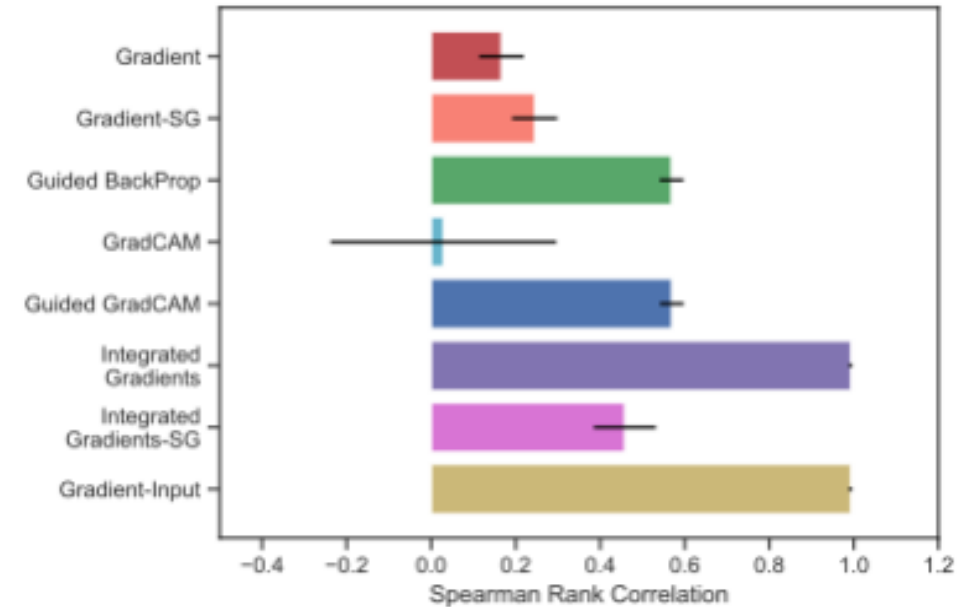
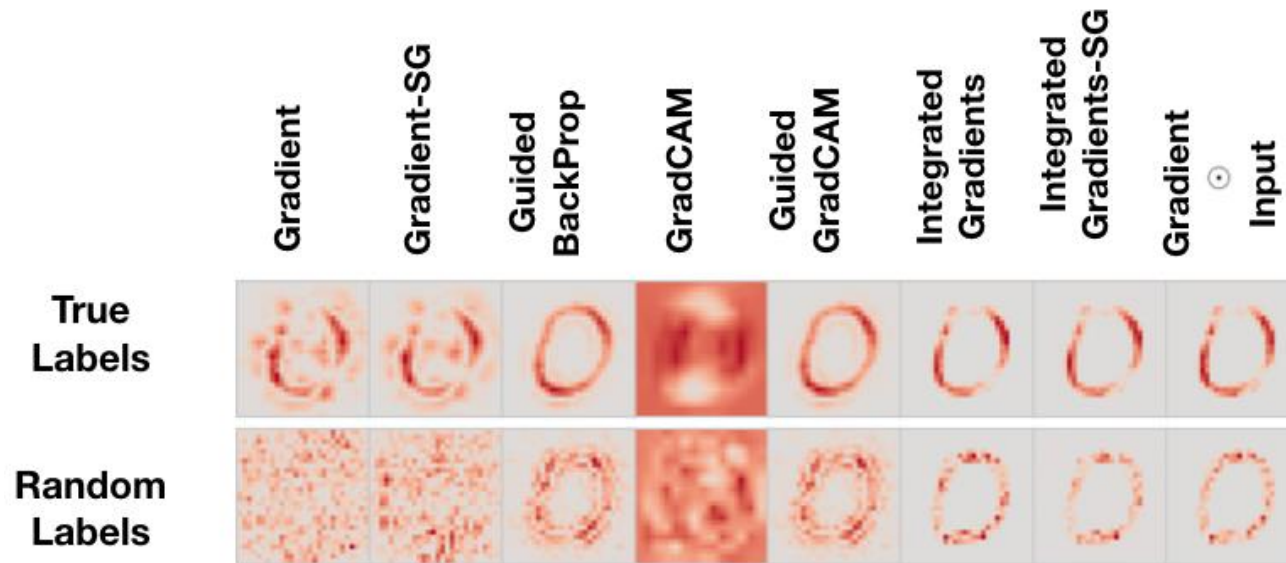
SANITY CHECKS

Model Parameter Randomization Test: "Randomizing layers should change explanations"



SANITY CHECKS

Data Randomization Test: "Randomizing labels should change explanations"



T H A N K S !

